RASSA Journal of Science for Society 4(1): 36-44, April 2022
DOI: 10.5958/2583-3715.2022.00005.3

Scientific Correspondence

Sugarcane Diseases: Diagnostics and Management

R. Viswanathan*

Division of Crop Protection, ICAR-Sugarcane Breeding Institute, Coimbatore-641007, Tamil Nadu, India

***Corresponding Author**
rasaviswanathan@yahoo.co.in

Key words
Red rot, smut, wilt, leaf scald, sugarcane mosaic, yellow leaf, foliar diseases

Received: January 12, 2022
Accepted: April 02, 2022

Sugarcane is a major field crop grown essentially for sugar and other sweeteners in India and in many other tropical and subtropical countries. Recently, it has emerged as a bioenergy crop for producing bioethanol for automobiles. Diseases caused by fungi, bacteria, viruses, phytoplasma and nematodes have become major constraints for sugarcane productivity. In spite of all the efforts of breeding for disease resistant varieties, this crop is becoming more and more prone to many diseases. The disease incidences are increasing at an alarming rate and the yield is falling down every year and sugarcane diseases cause a loss of several billions of dollars across the globe, every season. About 10-15 per cent of the nations sugar produced are lost due to diseases. Among the diseases red rot, smut, wilt and pineapple disease (sett rot) are the important fungal diseases. Bacterial diseases like leaf scald (LSD) and ratoon stunting (RSD) are found to cause considerable yield loss in some states. Also red stripe are known to cause damage in subtropical regions in India. Among the viral diseases, mosaic and yellow leaf (YLD) are the major diseases affecting sugarcane. Besides these, grassy shoot caused by phytoplasmas is also a potential disease, which can cause considerable damage to sugarcane production in all the states. Foliar diseases such as yellow spot, brown spot, brown stripe, eye spot, ring spot, rust etc may cause loss to sugarcane depending on the prevailing environmental conditions. Many promising varieties were removed from cultivation in the past since they succumbed to new pathogenic variants of red rot with more virulence. Also slow build up of many non-fungal diseases in sugarcane causes decline in varietal performance and results in varietal degeneration. In this review, important diseases of sugarcane and their management are briefly described.

MAJOR DISEASES AFFECTING SUGARCANE PRODUCTION IN INDIA

Red rot

It is one of the most widespread sugarcane diseases in the country and it has been a constraint for the past 100 years in India and other South Asian countries. It was involved in the failure of important commercial varieties like Co 312, Co 419, Co 453, Co 658, Co 997, Co 1148, Co 6304, Co 7805, CoC 671, CoC 85061, CoC 92061, CoJ 64, CoLk 8102, CoS 8436, SoSe 92423, CoSe 95422 etc (Viswanathan, 2021a). Currently the disease occurs in all the sugarcane growing states in India except Karnataka and Maharashtra states, more severely in the states of Uttar Pradesh and Bihar. The disease is caused by the fungal pathogen *Colletotrichum falcatum* Went. The pathogen infection seriously affects crop productivity since infected stalks become unfit for milling and partial infection also affect the quality of canes due to inversion of sucrose to reducing sugars. In the advanced stages, entire stool dries up. When such canes are split open longitudinally they show reddened areas with horizontal white patches. The entire parenchymatous tissues are affected and in later stages they form a hollow cavity with dark grey mycelial growth of the fungal pathogens. On the rind, dark brownish lesions of various sizes can be seen. Similarly necrosis on nodal region is also commonly observed. In such canes, cut ends show reddening with white patches.

Smut

The disease is caused by the fungus *Sporisorium scitamineum* (Syd.) M. Piepenbr., M. Stoll & Oberw. 2002 (Syn: *Ustilago scitaminea* H. & P. Sydow). The fungus has no alternate hosts. Disease affected stools give out excessive tillers which are lanky and end up with black whip structures. Affected clumps show profuse tillering with lanky tillers. Occasionally whips are produced from axillary buds also. The infected plants within a field often arise from planting systemically infected setts or from ratoons of infected clumps of previous crop. The disease is very well adapted to aerial dispersal and spread. Smut is generally favoured by hot dry weather conditions. High rainfall reduces the severity of smut development. Disease severity usually increases through ratoon crops. Plant stress increases frequency of whip development; cultivars that normally would not have whips may show symptoms under high stress.

Wilt

Fusarium sacchari (E.J. Butler) W. Gams is the disease causative fungus. The affected stools turn yellowish and finally dry as in the case of red rot. Longitudinal splitting of the affected cane shows dull brownish discolouration of internal tissue with linear pith cavities and desiccation. Wilt fungi are weak soil borne pathogens. Abiotic factors like drought, waterlogging, drought followed by water logging weaken the root system and predisposes the plant for wilt infection. Subterranean soil pests such as white grub, root borer and nematode and insect pests like mealy bug, scale insect, fungal pathogen like red rot weakens the plant and root system paving the way for wilt infection. However, the myth of root borer is always associated with wilt does not exist under field conditions (Viswanathan *et al.,* 2022). Wilt has caused significant losses in India where several epidemics have occurred. It is responsible for the elimination of many popular varieties from cultivation. Wilt is very common in certain locations where conducive environment and susceptible hosts are available (Viswanathan, 2013, 2020). In most of the delta regions and subtropical plains, endemic nature of the disease prevails. The disease adversely affects germination. Wilt incidence is always higher in ratoon crops compared with the plant crop. Combined infection of red rot and wilt causes more loss to the crop than their infection alone. Earlier it was thought that the disease occurs during maturity stage and its severity can be seen during harvest. It is now established that the disease occurs during germination onwards and cause death of the germinated settlings or young plants. Wilt infection in young crops can also be seen as dried canes with de-topped crown.

Sett rot (Pineapple disease)

Sett rot is caused by the fungal pathogen *Cerotocystis paradoxa.* The disease is common in germinating setts. The disease is primarily spread through soil borne inoculum under ill-drained conditions. The fungus infects the setts mainly through the cut ends and slowly spreads to the entire parenchyma tissues. Sett rot infected setts fail to germinate leaving many gaps in the field. In the early stage of rotting, a stinky odour of pineapple is emitted due to production of ethyl acetate and it helps in identifying the disease. The disease is also found to cause damage to standing canes in the field. Severe waterlogging, lodging of canes and other stresses favour disease build up in the field.

Ratoon stunting (RSD)

The disease is caused by the xylem limiting bacterium *Leifsonia* (*Clavibacter*) *xyli* subsp. *xyli* (Lxx) (Davis *et al.*). Diseased clumps usually display stunted growth, reduced tillering, thin stalks with shortened internodes and yellowish foliage. The characteristic stunting and unthriftness associated with RSD are usually greater when there has been a growing period with moisture stress. Growth of a disease crop is slower than that of a healthy crop, and ultimately the yield is reduced. The reduction in yield is due to the production of thinner and shorter stalks rather than a reduction in the total number of canes. The affected plants may not show any abnormality in the root system or in the underground buds and portions of the stalk. In general, the disease severity is more in older varieties. The term ‘ratoon stunting’ is misnomer since the disease also causes severe damage in plant crops. Diseased stalks of some varieties may exhibit an internal discolouration of vascular bundles at the lower portion of nodes, but these symptoms are often ephemeral. They appear as yellow to reddish brown dots, commas, or short lines when viewed by slicing longitudinally through nodes. The discolouration does not extend into the internode unlike similar symptoms due to other diseases (Viswanathan, 2012).

Grassy shoot

The disease is caused by sugarcane grassy shoot (SCGS) phytoplasmas. It occurs throughout the country and its severity is more in the areas where proper seed nursery programmes are not there. Infected stools show excessive and stunted tillers with narrow leaves. Most of the tillers do not develop into millable canes. In the affected clump, the few canes formed and are stunted with leaves being yellow or white. In some cases, axillary bud sprouting with yellow or paper white leaves may be seen. The whole infected stool appears like a big bunch of grass (Nithya *et al.,* 2020). Grassy shoot can cause very heavy yield losses particularly when planting material is obtained from infected sources. The disease severity will be more in ratoon crops and this contributes to poor yield in ratoons.

Mosaic

This is one of the oldest diseases recorded in India and is associated with varietal degeneration. *Sugarcane mosaic virus* (SCMV) and *Sugarcane streak mosaic virus* (SCSMV) cause mosaic disease in sugarcane in India. The virus has abundant variation and a single strain can infect sugarcane, maize and sorghum. Symptoms of mosaic may vary in intensity with cultivar, growing conditions, temperature and strain of the virus. The chlorotic areas are most easily seen in young, rapidly growing leaves, particularly near the basal portion of the leaf. The proportion of the leaf that is covered by the chlorotic areas may vary from scattered, short yellowish stripes to chlorotic areas that predominate over the leaf with islands of normal green. On older leaves, the symptoms tend to recover and appear as healthy. The virus causes mild mottling in the leaves with dark and light areas of chlorophyll development. With more virulent strains, stunting, yellowing, chlorosis and sometimes necrosis are also noticed. Recently we found that infection of debilitating strains affects yield and quality in sugarcane (Viswanathan and Balamuralikrishnan, 2005). Generally, long chlorotic streaks alternate with the normal green portions of the leaf are observed in different varieties in the country.

Yellow leaf (YLD)

This disease is caused by *Sugarcane yellow leaf virus* (ScYLV) and reported in India during 1999 (Viswanathan, 2002). The symptoms appear initially on matured leaves three through five usually in maturing plant or ratoon crop. The symptoms could be very clear after 5 to 6 months of crop growth. On the leaves, the symptom appears as yellowish midrib on the lower surface. The yellowing may be confined to midrib region or the yellow discolouration may spread laterally to adjoining laminar region parallel to midrib upto a distance of 2.0 cm. Reddish to pinkish discolouration of midrib and laminar region is also noticed in certain varieties. In most susceptible varieties, typical yellowing of midribs and laminar region is noticed on upper surface of the leaves. Finally symptoms of necrosis of discoloured laminar region from leaf tip to bottom and subsequent drying of entire leaf is noticed. In ratoon crop, the intensity of the disease will be much higher than in plant crop. The sugarcane varieties showing mild symptoms usually record normal cane growth. In severely infected clumps, cane thickness and stalk height are significantly affected. Severe infection of the disease leads to shortening of internodes in the top. This effect culminates in bunching of leaves at the top. Usually such infection results in drying of entire clumps. Severe infection of ScYLV leads to reduced juice quality and sugar recovery. Combined infection of ScYLV and RSD bacterium in sugarcane causes severe stunting than their infection alone (Viswanathan, 2002; 2004). The disease occurs in epidemic form on most of the cultivated varieties in India (Rao *et al.,* 2000) and it adversely affects cane productivity in the tropical region. The disease infection may result in 25-30 per cent loss to cane and juice yield in popular varieties (Viswanathan, 2012, 2021c). Further, impact of the virus infection on physiological parameters viz. photosynthetic rate (A), stomatal conductance (gs) and SPAD meter values and growth/yield parameters, such as stalk height, stalk thickness and number of internodes revealed significant reduction in sugarcane cultivars. In addition, to reductions in stalk weight, height and girth, YL also reduced juice yield in the affected canes up to 34.15% (Viswanathan *et al.,* 2014a).

Varietal degeneration

Sugarcane pathogens like SCMV, SCSMV, ScYLV, SCGS-phytoplasmas and ratoon stunting bacterium systemically infect sugarcane and over the years the varieties degenerate due to the systemic colonization of these pathogens (Bagyalakshmi *et al.,* 2019; Viswanathan, 2012, 2013, 2016, 2018). Decline in varietal performance over the years is mainly due to accumulated pathogens inside the stalk affecting cane growth and photosynthetic efficiency, which directly results in reduced cane yield and sugar yield. Although these viral/ bacterial pathogens cause limited symptoms in the field, continuous vegetative propagation results in enhanced pathogen load that would increase the pathogenic potential to cause severe disease symptoms. Combined infection of two or more viral/ bacterial pathogens accelerates the damage to the crop in the field and this is due to infection of one pathogen makes the plant more susceptible to another. In this way, a variety degenerates faster and its potential comes down over the years. Author has witnessed such degeneration in a popular cv Co 419 in Karnataka state due its high susceptibility to mosaic, YLD and RSD. Similarly, the cv CoC 671 another popular variety of tropical region degenerated due its high susceptibility to mosaic and YLD in different parts of Karnataka and Maharashtra. In the subtropical region, CoS 767, the popular variety recently adversely suffered from varietal degeneration caused by the pathogens causing SCGS, RSD, YLD and mosaic and it led to its replacement by the elite variety Co 0238 and it became very popular in the region.

Pokkah boeng

The disease is common during rainy months in the field. Although under normal situations it may not cause significant yield loss it has the potential to arrest the crop growth temporarily. The disease occurs throughout the world and severe forms of the disease are recorded in high humidity areas. *Fusarium verticillioides* and *F. sacchari* are the causative fungi (Teleomorph: *Gibberella fujikuroi* (K. Sawada) H.W. Wollenweber). Recently association of *F. proliferatum* with the disease has been found in India. The disease manifests in two phases viz. *pokkah boeng* (PB) and top rot. The most common symptom is a malformed or twisted top, which gives this disease its name "*pokkah boeng*" from the Javanese language. Symptoms develop during rainy periods which coincide with grand growth period. Initially, young leaves are chlorotic at their base and patchy elsewhere on the blade. Chlorosis is most obvious on the lower surface of the leaf or in twisted laminar regions, where white mycelium may be seen. Affected leaves tend to be narrow at the base. Development of further symptoms is dependent on the susceptibility of the variety and on environmental conditions conducive to the pathogen. Young leaves may become infected in the spindle, resulting in pronounced wrinkling, twisting and shortening of the leaves. Sometimes, the leaves are shortened to few inches without lamina having malformed midrib or growth of the leaves ceased to few inches without malformation giving a de-topped spindle. As the leaves mature, irregular reddish stripes and specks develop within the chlorotic areas. Infection in the spindle may reach the growing point and continue into the stalk (Viswanathan, 2012; Viswanathan *et al.,* 2014b). Upon recovery from the disease, varying intensities of retardation of internode growth, bud sprouting and 'knife-cut' symptoms on the stalks are observed.

Sometimes, the growing point is killed leading to development of top rot. Due to death of spindle, sprouting of the lateral buds occurs. Most of the PB-affected canes generally recover from the symptoms but in cases of top rot recovery is not there. Upon recovery we notice the normal whorl with remnants of twisted leaf portions of affected leaves still twisting around the spindle. This disease is favoured by warm, moist growing conditions. Symptom development begins early in the rainy season which normally coincides with rapid and vigorous growth of the canes. The three to seven months-old are most susceptible to the disease. Conidia are air borne and are deposited on plants, then washed by rain into infection sites. Recently we found that the diseases wilt and PB are caused by the same fungal pathogen *F. sacchari* (Viswanathan *et al.,* 2017). This has added a new finding in epidemiology of the two diseases caused by *Fusarium*. It was demonstrated that stalk infection of *F. sacchari* led to foliar infections and vice versa. Such epidemiological behavior of the pathogen is also attributed to sudden emergence of PB in different states.

Leaf fleck

This is a virus disease, lesser known to personnel in sugar industry and sugarcane scientists. The disease is caused by *Sugarcane bacilliform virus* (SCBV) and occurs throughout the country. Earlier, the disease was thought to be confined to sugarcane germplasm but detailed studies of the author evidenced virus infection in many popular varieties under field conditions in the country. Major symptoms of the disease include appearance of flecks or specks throughout leaf lamina. However, the flecks are more prominently seen on leaf margins and distal end of the leaf. Gradually, the chlorotic flecks increase in size, turn yellow and reddish. In extremely severe cases entire leaf turns yellow or reddish and symptoms appear close to yellow spot symptoms. Severe disease expression leads to premature drying and death of the leaves (Viswanathan *et al.,* 2019). Such disease severity is noticed in east coast region in the popular varieties such as CoA 92081 (87A298), CoV 09356 (2003V46), CoV 92102 (83V15) etc.

Foliar diseases

The foliar diseases such as rust, eye spot, yellow spot, brown stripe and ring spot are air borne and most of them survive on other collateral and weed hosts. None of the foliar pathogens are sett-borne. High moisture or relative humidity following rains accompanied by low or cool temperatures favours their incidence. During this period, excess irrigation and non-stripping off their lower leaves and dry leaves, leading to high relative humidity build up within the crop. Such microclimatic conditions help to build up of the disease to epidemic levels. Among the different foliar diseases, rust has become more severe and occurs in epidemic form in Maharashtra and Karnataka during post monsoon season. Almost all the varieties under cultivation such as CoM 0265, CoVSI 9805, Co 86032, Co 92005, CoC 671, Co 94012, etc were affected by rust. The newly introduced variety CoM 0265 shows high susceptibility to brown spot in Maharashtra and Karnataka.

Integrated disease management

The various types of diseases on sugarcane determine the quality, quality and stability of crop yield. This long duration cash crop due to its vegetative propagation, high sugar accumulation and practice of ratooning makes it easily susceptible to the diseases in the field. Besides this, wide spread practice of monoculture and heavy wet condition add to its susceptibility. This unfavourable environment in host plant invites large number of pathogens. The pathogens are not only reducing the yield but also cause the deterioration of the variety due to their accumulation in the stalk over the time. This phenomenon is referred to as 'varietal degeneration' and this results in loss of full potential of a variety and subsequently such varieties are withdrawn from cultivation. In the past and even today many high yielding, high sugar and popular cane cultivar like Co 419, Co 740, CoC 671 etc are being withdrawn from cultivation only because of their high susceptibility to red rot, smut, grassy shoot and mosaic. No single method is efficient / available to control sugarcane diseases due to various reasons hence an integrated approach involving cultural, chemical/physical methods, host resistance and legislative measures is suggested for the sustainable management of sugarcane diseases (Viswanathan, 2012).

Infected planting materials are responsible for the primary spread of the disease in the field. Hence, going for the disease-free setts would reduce the risk of disease introduction to disease free areas. Lack of awareness on seed cane health and ignoring quarantine regulations resulted in introduction of diseases, their epidemics and varietal degeneration in the country. To increase sugarcane productivity, supply of healthy seed canes is to be ensured in the field. As vegetative propagation in sugarcane favours harbouring of the pathogens causing red rot, smut, wilt, SCGS, LSD, YLD and RSD in the setts, adequate care should be taken while selecting seed canes. Since it is difficult to detect incipient infections of *C. falcatum* in seed-pieces, it is recommended to take the planting material from a disease free crop. Any crop with more than 5% smut or as high as 2% grassy shoot incidence is unsuitable for seed purpose. For red rot, if there is any infected clump in the field, the plot is to be rejected for seed. It is advised to select always a disease free area to raise the seed crop.

Disease surveillance and diagnosis

Many of the diseases do not cause diagnosable symptoms on the seed cane and different factors influence disease expression in the field hence we need to follow certain diagnostic techniques based on serology or molecular biology to detect them in the seed cane before planting in the field. Effort to detect and diagnose sugarcane pathogens using more advanced laboratory techniques has been achieved in the past two decades. Detailed studies conducted over the years at ICAR-SBI, Coimbatore resulted in standardizing molecular diagnosis of different pathogens infecting sugarcane. Currently, PCR assay is used to index sugarcane materials for SCGS infection and RT-PCR technique is used to index sugarcane for SCMV, SCSMV and ScYLV infections. These diagnostics tests have become imperative to raise disease-free planting materials. Hence tissue culture production units in the country are utilizing the indexing service from our lab, which had received accreditation for virus indexing from Department of Biotechnology, GoI during 2008. The molecular tests are highly sensitive to detect very low virus titre in *in vitro* stock culture or in seedlings.

Since association of two viruses either alone or in combination in causing mosaic in sugarcane, duplex (D-) RT-PCR) assays were developed to detect the associated viruses in a single reaction (Viswanathan *et al.,* 2008). Further, the D-RT-PCR was found equally reliable to uniplex RT-PCR performed with SCMV and SCSMV primers in separate reactions. Subsequently, a multiplex-RT-PCR was developed for the detection of SCMV, SCSMV and ScYLV, three of the major RNA viruses widely prevailing in the sugarcane growing regions around the world (Viswanathan *et al.,* 2010). Polyclonal antisera were produced against recombinant SCSMV coat protein (CP) and in direct antigen coating (DAC)-ELISA assays, the

SCSMV-cp antiserum is highly efficient to detect SCSMV infection in naturally infected / asymptomatic sugarcane field samples (Viswanathan *et al.,* 2013a). Further, in immune-capture (IC)-RT-PCR assays, the same recombinant antiserum trapped both SCSMV and SCMV and it was sensitive enough to detect the two viruses causing mosaic (Viswanathan *et al.,* 2013b). Although DAC-ELISA is simple and cost effective to diagnose these viruses, in samples with very low virus titre, IC-RT-PCR would be more sensitive. Recently RT-loop-mediated isothermal amplification (RT-LAMP) based assays were developed to detect sugarcane viruses in the lab without conventional PCR reactions (Anandakumar *et al.,* 2018, 2020). Lateral flow immunochromatographic assays (LFIAs) are simple, rapid and portable detection now popular in biomedicine, food, environment, and agriculture. Taking advantages of the new LFIA technology, we developed LFIA for SCMV and SCSMV detection with a high sensitivity. The gold nanoparticle-based LFIA assay was found to be economical and affordable for farmers and is also a practically feasible diagnostic assay for diagnosing sugarcane viruses (Muthuramalingam *et al.,* 2022).

Ratoon management

We notice more damages caused by the diseases in ratoons than plant crop due to different reasons. Inoculum level of systemic pathogens causing smut, grassy shoot, LSD, RSD, mosaic etc. gradually increases and results in severe expression of diseases in a ratoon crop. Combined infection of two diseases such as RSD and mosaic or RSD and YLD adversely affects the crop growth and such effect is more pronounced in ratoons. Also establishment of ratoon crop in the field is severely affected by pathogen infection in plant crops. Similarly, more accumulation of pathogen in ratoons facilitates acquiring higher virulence among the pathogens. The popular sugarcane variety Co 419 lost its prominence due to its susceptibility to RSD, mosaic and YLD. Higher RSD pathogen load favours YLD and entire foliage become yellowish and slowly crop degenerates and dries. This situation was found more in ratoons, due to high pathogen load/vigour. Since multi-ratooning of sugarcane has several advantages, it is being followed in many countries. However, to sustain sugarcane productivity and to improve ratoon productivity these points are to be given due consideration.

Emerging new diseases

Recently many diseases have emerged as serious ones in different countries. YL disease has spread to epidemic form in some varieties. Although mild infections of the disease do not cause much crop loss continuous use of seed from such fields lead to severe disease out break. Similarly, problems of wilt, RSD and SCGS have been ignored in certain pockets, as there was no previous history of their severity there. Due to introduction of new varieties or change in cultivation pattern, minor diseases become severe. Hence, we have to keep a constant vigil on the emerging diseases in the region and immediately suitable management practices are to be taken up to avert future losses.

Chemical control

Chemical controls are possible for few of the diseases, particularly those caused by fungal pathogens. Sett rot pathogen survives in the soil. As a prophylactic measure, the setts are to be dipped in fungicide solution to protect the cut-ends from the pathogen. Recent studies at the Institute revealed that sett treatment of thiophanate methyl fungicide in combination with biocontrol bacterium *Pseudomonas* reduces soil borne infection of red rot pathogen surviving in debris. Similarly in case of smut, propiconazole was found effective and in ratoon crops. Two sprays immediately after sprouting reduce the disease build up in the ratoons. If rust is severe, five to six sprayings of mancozeb (0.2%) between November and March is recommended to control the disease under our conditions. Similarly to control eye spot spraying of copper oxy chloride or Mancozeb (0.2%) once in 30 days during initiation period is recommended. Whenever the disease is high, fungicidal application should be sprayed at 18-20 day intervals. Recently detailed studies conducted at the institute on effective delivery of fungicides in single bud or two budded setts/ bud chips utilizing mechanized-vacuum infiltration approach and the treatment has resulted in more effective diffusion of the chemical into sugarcane setts / buds due to reduced pressure created in the treatment chamber. The newly devised sett treatment device is portable and simple to operate. Recycling of chemicals resulted in huge savings in chemicals used for pre-treatment, thus making the system environmental friendly. Field trials conducted at Coimbatore and red rot and smut endemic locations revealed that fungicide delivery through the new device efficiently protected the crop from these diseases. In addition, delivery of different agrochemicals for sugarcane multiplication nurseries (bud-chip or single bud) can be delivered through the device (Malathi *et al.,* 2017; Viswanathan *et al.,* 2016, 2017).

Use of resistant varieties

The use of resistant varieties is the most important means of managing sugarcane diseases in a sustainable way. Although frequent breakdown of sugarcane varieties is a cause of alarm, new varieties with high to moderate levels of resistance to the disease are introduced time to time and saved the crop from severe red rot epiphytotics (Viswanathan, 2021b). Unlike red rot, smut resistance in commercial varieties in the field is quite stable. Most of the ruling varieties are resistant to smut. While advocating sugarcane varieties to a factory care should be taken that there should be a proper varietal admixture in that region. In case of disease outbreaks, the disease spread will be rapid in case of monocropping as there is no barrier to check the disease. Also cultivating a single variety in more than 50 per cent of the area would result in development of highly virulent strain of the pathogen, which may cause more damage to the crop than the existing pathotypes.

Heat therapy

The scientific principle involved in heat therapy is that the pathogens present in seed materials are inactivated or eliminated at set temperatures not deleterious for the host tissues. Aerated stream therapy (AST) or moist hot air treatment (MHAT) is advocated to eliminate sett borne infections of SCGS and RSD. Any fault in the AST or MHAT unit may adversely affect the sett germination. Hence functioning of the heating unit and temperature control systems, proper volume and circulation of the heating medium and proper loading of the cane within the treatment chamber are to be monitored time to time. Treated setts should always be treated with fungicides (carbendazim 0.1%) to reduce the entry of soil pathogens through cut ends. In place of the thermotherapy, meristem-culture derived seedlings can be used in three tier seed nursery programme to get disease-free seedlings.

Virus elimination

Since many of the viruses are systemically infecting sugarcane virus elimination through meristem-tip culture is being followed in many countries. *In vitro* culture techniques employed for virus elimination involve indirect morphogenesis. Production of ScYLV-free seedlings has ensured supply of YL-free planting materials to the growers fields and such fields showed renewed vigour in the crop. Overall, virus elimination through meristem culture combined with molecular diagnosis has been demonstrated as a viable strategy to manage YLD, which occurred in epidemic form in sugarcane in the recent years. Utilization of these molecular techniques will be a boon to raise disease free planting materials for sugarcane plantations. This also aids maintaining the popular varieties for many years without degeneration to sustain higher productivity. Recently we have demonstrated that planting of YL-free planting materials gives cane yield of 250 tonnes per hectare in the popular variety Co 86032 in Erode Dt, Tamil Nadu (Viswanathan *et al.,* 2018).

CONCLUSION AND FUTURE PERSPECTIVES

The fungal diseases like red rot, smut and wilt were responsible for the elimination of many elite commercial varieties in the past in different epidemics. Additionally many of the non-fungal diseases contribute to decline in their performance due to 'varietal degeneration'. Lack of awareness on seed cane health and ignoring quarantine regulations resulted in introduction of diseases, their epidemics and varietal degeneration in the country. Hence sugar industry may follow strict quarantine norms when they bring new varieties from unknown sources or destinations. This would prevent introduction of new diseases which are not reported in the region from other zones or countries. Growing disease resistant varieties is the mainstay in disease management in sugarcane and introduction of disease resistant varieties after red rot or smut epidemics has saved sugar industry during several occasions. Although YLD has created a havoc to sugarcane cultivation in the country and SBI has evolved strategies to manage the disease through meristem culture combined with molecular diagnosis of the virus. Hence, need of the hour is to establish YLD-free nurseries in different sugar mills to reduce disease severity. Wherever appropriate, conventional seed nursery programme through aerated steam therapy needs to be followed to reduce severity of grassy shoot and RSD. Disease surveillance programmes in the country need further strengthening including use of remote sensing approach and usage of drones. This would lead to creation of disease maps for various diseases in sugarcane and this would facilitate developing possible forewarning systems and varietal deployment in a region in the future. We witnessed outbreak of *pokkah boeng* and rust recently in different parts of the country hence there is a need to create awareness about the potential diseases which may seriously affect sugarcane production. Hence the sugar industry should be prepared to tackle minor diseases when they become major diseases in sugarcane.

REFERENCES

Anandakumar L, Bagyalakshmi K, Muthuramalingam TR, Nithya K, Parameswari B and Viswanathan R (2020). Reverse transcription loop-mediated isothermal amplification based rapid detection of *Sugarcane mosaic virus* and *Sugarcane streak mosaic virus* associated with mosaic disease of sugarcane. *Ind. Phytopathol.,* 73: 349-358. doi: 10.1007/s42360-020-00219-w

Anandakumar L, Bagyalakshmi K, Nithya K, Parameswari B and Viswanathan R (2018). Reverse transcription loop-mediated isothermal amplification (RT-LAMP) assay for rapid diagnosis of Sugarcane yellow leaf virus in sugarcane. *Sugar Tech,* 20: 708-716. doi: 10.1007/s12355-018-0606-2

Bagyalakshmi K, Viswanathan R and Ravichandran V (2019). Impact of the viruses associated with mosaic and yellow leaf disease on varietal degeneration in sugarcane. *Phytoparasitica,* 47: 591–604.

Malathi P, Viswanathan R and Sundar AR (2017). Mechanized means of sett treatment: an effective way of delivering fungicides for the management of red rot in sugarcane. *Sugar Tech,* 19: 176-182.

Muthuramalingam TR, Nithya K, Parameswari B and Viswanathan R (2022). Ultrasensitive nano-gold labelled, duplex lateral flow immunochromatographic assay for early detection of sugarcane mosaic viruses. *Sci. Rep.,* 12: 4144. https://doi.org/10.1038/s41598-022-07950-6

Nithya K, Parameswari B, Bertaccini A, Rao GP and Viswanathan R (2020). Grassy shoot: the destructive disease of sugarcane. *Phytopathogenic Mollicutes,* 10(1): 10-24.

Rao GP, Gaur RK, Singh M, Srivastava AK, Virk KS, Singh N, Viswanathan R, Patil AS and Jain RK (2000). Occurrence of sugarcane yellow leaf virus in India. *Sugar Tech,* 2(4): 37-38.

Viswanathan R (2002). Sugarcane yellow leaf syndrome in India: Incidence and effect on yield parameters. *Sugar Cane. Intern.,* 5: 17-23.

Viswanathan R (2004). Ratoon stunting disease infection favours severity of yellow leaf syndrome caused by sugarcane yellow leaf virus in sugarcane. *Sugar Cane. Intern.,* 22(2): 3-7.

Viswanathan R (2012). Sugarcane diseases and their management. ICAR-Sugarcane Breeding Institute, Coimbatore, India, p140.

Viswanathan R (2013). Status of sugarcane wilt: One hundred years after its occurrence in India. *J. Sugarcane Res.,* 3: 86–106.

Viswanathan R (2016). Varietal degeneration in sugarcane and its management in India. *Sugar Tech,* 18: 1–7.

Viswanathan R (2020). Fusarium diseases affecting sugarcane production in India. *Indian Phytopathol.,* 73: 415–424. https://doi.org/10.1007/s42360-020-00241-y

Viswanathan R (2021a). Red rot of sugarcane (*Colletotrichum falcatum* Went). *CAB Reviews,* 16, No. 023, doi: 10.1079/PAVSNNR202116023

Viswanathan R (2021b). Sustainable sugarcane cultivation in India through threats of red rot by varietal management. *Sugar Tech,* 23: 239–253.

Viswanathan R, Balaji CG, Selvakumar R, Malathi P and Sundar AR (2017). Epidemiology of *Fusarium* diseases in sugarcane: a new discovery of same *Fusarium sacchari* causing two distinct diseases, wilt and Pokkah Boeng. *Sugar Tech,* 19: 638-646.

Viswanathan R and Balamuralikrishnan M (2005). Impact of mosaic infection on growth and yield of sugarcane. *Sugar Tech,* 7: 61–65.

Viswanathan R, Balamuralikrishnan M and Karuppaiah R (2008). Duplex - reverse transcription - polymerase chain reaction (D-RT-PCR) - a technique for the simultaneous detection of viruses causing sugarcane mosaic. *Sugar Tech,* 10: 81-86.

Viswanathan R, Chinnaraja C, Malathi P, Gomathi R, Rakkiyappan P, Neelamathi D and Ravichandran V (2014). Impact of sugarcane yellow leaf virus (ScYLV) infection on physiological efficiency and growth parameters of sugarcane under tropical climatic conditions in India. *Acta Physiol. Plant,* 36: 1805–1822.

Viswanathan R, Karuppaiah R, Bagyalakshmi K, Balan S and Kaverinathan K (2019). Emergence of leaf fleck caused by *Sugarcane bacilliform virus* in sugarcane as a serious disease under field conditions in India. *Intern Sugar J.,* 121: 146-153.

Viswanathan R, Malathi P, Annadurai A, Prasanth CN and Scindiya M (2014c). Sudden occurrence of wilt and pokkah boeng in sugarcane and status of resistance in the parental clones in national hybridization garden to these diseases. *J. Sugarcane Res.,* 4(1): 62–81.

Viswanathan R, Malathi P and Naik R (2016). A sett treatment device for fungal disease management and healthy nursery programme in sugarcane. *Proc. Ann. Con. Sugar Technol. India,* 77: 236-253.

Viswanathan R, Malathi P and Neelamathi D (2018). Enhancing sugarcane yield per hectare through improved virus-free seed nursery programme. *ICAR NEWS,* 24(4): 4-5.

Viswanathan R, Malathi P, Prasanth CN, Naik R and Annamalai SJK (2017). Sett way of growing sugarcane for fungal treatment. *ICAR NEWS,* 23(2): 1-3.

Viswanathan R, Selvakumar R, Geetha N, Balaji CG, Annadurai A, Pazhani AS, Malathi P., Sundar AR, Nithiyanantham R and Manivannan K (2022).

Epidemiology of sugarcane wilt: predisposition by root *borer Polyocha depresella* a myth or reality. *Ind. Phytopathol.,* 75: 3-14.

Viswanathan R (2018). Changing scenario of sugarcane diseases in India since introduction of hybrid cane varieties: path travelled for a century. *J. Sugarcane Res.,* 8(1): 1–35.

Viswanathan, R, Karuppaiah, R and Balamuralikrishnan M (2010). Detection of three major RNA viruses infecting sugarcane by multiplex reverse transcription–polymerase chain reaction (multiplex-RT-PCR). *Austral Plant Pathol.,* 39: 1–6.

Viswanathan, R., Ganesh Kumar V, Karuppaiah R, Scindiya M and Chinnaraja C (2013a). Diagnosis of *Sugarcane streak mosaic virus* using recombinant coat protein antiserum in sugarcane. *Int. Sugar J.,* 115: 279-283.

Viswanathan, R., Ganesh Kumar V, Karuppaiah R, Scindiya M and Chinnaraja C (2013b). Development of duplex-immunocapture (duplex-IC) RT-PCR for the detection of *Sugarcane streak mosaic virus* and *Sugarcane mosaic virus* in Sugarcane. *Sugar Tech,* 15: 399–405.

Viswanathan R (2021c). Impact of yellow leaf disease in sugarcane and its successful disease management to sustain crop production. *Indian Phytopathol.,* 74: 573–586 DOI:10.1007/s42360-021-00391-7